

Jennifer Chang, Ph.D.

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SUMMARY **Bioinformatics and workflow developer** with 8+ years of experience.

EDUCATION Ph.D. in **Bioinformatics and Computational Biology** *Aug 2011 – June 2017*
minor in **Statistics**, dissertation on "Designing an integrated system for biological network exploration"
Iowa State University, Ames, Iowa 50010, USA

B.A. in **Computer Science** and **Biochemistry** *Aug 2007 – May 2011*
Cornell College, Mount Vernon, Iowa 52314, USA

PROFESSIONAL EXPERIENCE **Senior Bioinformatic Analyst** St. Jude Children's Research Hospital *Feb 2026 – present*

- Part of the Microbial Computational Genomics Collaborative Center with Dr Chrispin Chaguza, to support the Host Microbe Interaction Department
- Discuss, design, implement, deploy, debug, diagram, and document pipelines written in a combination of bash, R, python, nextflow, wdl (sprocket!) to run on the the local HPC (LSF or SLURM as needed)
- Collaborate across St Jude Campus by actively participating in Data Science in AI, and Informal Bioinformatics Meetings
- Broad level discussions of viral/microbe genomics and biological questions
- 40 hrs/wk. Supervisor: Chrispin Chaguza

Bioinformatics Analyst III (Nextstrain) Fred Hutch Cancer Center *Nov 2021 – Feb 2026*

- Remote Contractor from November 2021 to September 2022, after which I became a full-time employee
- Delivered 300+ GitHub PRs across Nextstrain tooling and pathogen-specific repositories, improving performance, modularity, and long-term maintainability.
- Developed and maintained Snakemake workflows, with support in WDL (Terra.bio) and Nextflow as needed, enabling scalable and reproducible genomic analysis.
- Balanced standardization of heterogeneous pathogen pipelines under the pathogen-repo-guide with bespoke automation for dengue, Lassa, WNV, mumps, and norovirus pipelines summarizing trends.
- Ran harmonization of HI titer data integrating antigenic data with phylogenetics for seasonal influenza surveillance reports and global projections.
- Delivered a Nextstrain workshop at VEME 2023 (South Africa), training 50+ researchers on real-time phylogenetics analysis.
- Co-authored collaborative publications advancing open genomic surveillance.
- 40 hrs/wk. Supervisors: Trevor Bedford, Jover Lee

Bioinformatics Post-Doc, ORISE SCINet Fellow USDA-ARS *Jun 2020 – Nov 2021*

- Developed Nextflow pipelines for GATK (Hufford Lab) and VGP Assembly (ag100pest) projects, improving automation, documentation, and reproducibility across collaborative genomic projects
- Founded and organized SCINet Corner, a monthly virtual HPC knowledge-sharing series engaging participants across USDA locations
- Co-authored and maintained tutorials in the Bioinformatic Workbook (e.g. WGCNA), increasing accessibility of analytical methods
- Contributed analyses that supported two peer-reviewed genomic publications
- 40 hrs/wk. Supervisors: Andrew Severin, Brian Scheffler

- Performed large-scale statistical and genomic analysis (R, Bash, and Perl) of influenza A viruses (IAV) in swine, supporting national surveillance efforts, with a particular emphasis on reassortment.
- Collaborated with experimental and computational teams to design studies for swine or computation
- Produced quarterly IAV surveillance reports using Sweave, presenting trends and actionable findings to multi-site scientific teams.
- Selected a curated monthly set of 21 virus for whole genome sequences and deposition into NCBI GenBank, supporting public data availability.
- 40 hrs/wk. Supervisor: Amy Baker

Co-Founder Complex Computation, LLC; Ames, Iowa, USA

July 2015 – August 2017

- Co-developer of Mango Graph Studio, a C++ software for massive heterogeneous multi-network analysis
- Co-Founder leading efforts for the company to provide software solutions and workshops on network analysis.
- Served as Principal Investigator on two DARPA SBIR Contracts (W911NF-15-P-0040 in 2015 and W911NF-17-P-0014 in 2016), delivering monthly technical reports and providing findings to program managers.
- Presented the findings in-person to DARPA at a conference in Durham, NC.
- Collaborated with scientists and researchers at Iowa State University and other locations to develop network analysis pipelines to answer a variety of biological questions.
- Varied from 20 to 40 hours/week. Supervisor: Hui-Hsien Chou

Research Assistant Complex Computational Laboratory; Iowa State Univ.

Feb 2012 – Aug 2017

- Modernized the Lucy2 C++ wxWidgets GUI (2013), increasing usability and contributing to the subsequent 700+ cross-platform (Mac, Windows, Linux) downloads.
- Developed Mango Graph studio, a multi-network visualization and exploration software with a custom graph exploration language (Gel); later commercialized by licensing to Complex Computation, LLC.
- Awarded the Plant Sciences Institute Scholar Grant (2015) and received multiple conference awards for software and research presentations or posters.
- 20 hrs/wk. Supervisor: Hui-Hsien Chou

SELECTED
PUBLICATIONS

Huddleston, J., Bedford, T., **Chang, J.**, Lee, J., and Neher, R. A., 2024. Seasonal influenza circulation patterns and projections for February 2024 to February 2025. Zenodo. <https://doi.org/10.5281/zenodo.10846007>

Chang, J., Stahlke, A.R., Chudalayandi, S., Rosen, B.D., Childers, A.K. and Severin, A.J., 2023. polish-CLR: a Nextflow workflow for polishing PacBio CLR genome assemblies. *Genome Biology and Evolution*, 15(3), p.evad020.

Chang, J., Anderson, TK., Zeller, MA., Gauger, PC., Vincent, AL., "OctoFLU: Automated classification to evolutionary origin of influenza A virus gene sequences detected in U.S. swine," 2019 bioRxiv: ASM Journal Microbiology Resource Announcements (MRA)

Duwell, M., Feldman, K., Blythe, D., Radebaugh, M., Kough, E., Bachaus, B., Crum, D., Perkins, K., Blanton, L., Davis, C., Jang, Y., Vincent, A., **Chang, J.**, Abney, D., Gudmundson, L., Brewster, M., Polsky, L., Rose, D., "Influenza A(H3N2) Variant Virus Outbreak at Three Fairs – Maryland, 2017", *MMWR Morb Mortal Wkly Rep* 2018;67:1167-1173. DOI: <http://dx.doi.org/10.15585/mmwr.mm6742a1>

Chang, J., Cho, H., and Chou, H., "Mango: combining and analyzing heterogeneous biological networks", *BioData Mining*, August 2016

Andrews, K.R., **Chang, J.**, Roemer, C., Hadfield, J., Lin, V., Brito, A.F., Daodu, R., Joia, I.A., Kistler, K., Li, A.W., Moncla, L.H., Paredes, M.I., Kühnert, D., Torres, L.M., Voith, L., Aksamentov, I., Hodcroft, E.B., Huddleston, J., McCrone, J.T., Anderson, J.S.J., Sibley, T.R., Lee, J., Neher, R.A. and Bedford, T. Nextstrain automates real-time phylogenetic analysis of open data for endemic and emerging pathogens. 2026 bioRxiv. doi: 10.64898/2026.03.23.713807.

Daodu, R.O., **Chang, J.**, Prescott, J., Reinert, K., and Kühnert, D. Lassa virus live tracking and lineage assignment: how Nextstrain can enhance surveillance and public health in Africa and beyond. 2026 Emerging Microbes and Infections, p.2640699.

Janzen, G.M., Inderski, B.T., **Chang, J.**, Arendsee, Z.W., Janas-Martindale, A., Torchetti, M.K., Baker, A.L. and Anderson, T.K. Sources and sinks of influenza A virus genomic diversity in swine from 2009 to 2022 in the United States. 2025 Journal of Virology, pp.JVI-00541-25.

Zeller M.A., **Chang J.**, Trevisan G, Gauger PC, Zhang J. Nextclade data set for the ORF5-based lineage classification of PRRSV-1. Microbiology resource announcements. 0:e00303-25.

Zeller, M.A., **Chang, J.**, Trevisan, G., Main, R., Gauger, P.C. and Zhang, J., Rapid PRRSV-2 ORF5-based lineage classification using Nextclade. Frontiers in Veterinary Science, 11, p.1419340.

Huddleston, J., Bedford, T., **Chang, J.**, Lee, J., and Neher, R. A., 2024. Seasonal influenza circulation patterns and projections for February 2024 to February 2025. Zenodo. <https://doi.org/10.5281/zenodo.10846007>

Chang, J., Stahlke, A.R., Chudalayandi, S., Rosen, B.D., Childers, A.K. and Severin, A.J., 2023. polish-CLR: a Nextflow workflow for polishing PacBio CLR genome assemblies. Genome Biology and Evolution, 15(3), p.evad020.

Stahlke, A.R., **Chang J.**, Chudalayandi, S., Heu, C.C., Geib, S.M., Scheffler, B.E., Childers, A.K. and Fabrick, J.A., 2023. Chromosome-scale genome assembly of the pink bollworm, *Pectinophora gossypiella*, a global pest of cotton. G3: Genes, Genomes, Genetics, 13(4), p.jkad040.

Venkatesh D, Anderson TK, Kimble JB, **Chang J**, Lopes S, Souza CK, Pekosz A, Shaw-Saliba K, Rothman RE, Chen KF, Lewis NS., Baker ALV, Antigenic characterization and pandemic risk assessment of North American H1 influenza A viruses circulating in swine. Microbiology Spectrum. 2022 Nov 1:e01781-22.

Arendsee, Z., **Chang, J.** and Wurtele, E.S., 2022. rmonad: pipelines you can compute on. R Journal, 14(1).

Stahlke, A.R., **Chang, J.**, Tembrock, L.R., Sim, S.B., Chudalayandi, S., Geib, S.M., Scheffler, B.R., Perera, O.P., Gilligan, T.M., Childers, A.K. and Hackett, K.J., A chromosome-scale genome assembly of a *Bacillus thuringiensis* Cry1Ac insecticidal protein resistant strain of *Helicoverpa zea*. 2022 Genome Biology and Evolution.

Souza, C.K., Anderson, T.K., **Chang, J.**, Venkatesh, D., Lewis, N.S., Pekosz, A., Shaw-Saliba, K., Rothman, R.E., Chen, K.F. and Vincent, A.L., Antigenic distance between North American swine and human seasonal H3N2 influenza A viruses as an indication of zoonotic risk to humans. 2022 Journal of Virology, pp.JVI-01374.

Arendsee, Z.W., **Chang, J.**, Hufnagel, D.E., Markin, A., Janas-Martindale, A., Vincent, A.L. and Anderson, T.K., octoFLUshow: an Interactive Tool Describing Spatial and Temporal Trends in the Genetic Diversity of Influenza A Virus in US Swine. 2021 Microbiology Resource Announcements, 10(50), pp.e01081-21.

Kaplan, B.S., Anderson, T.K., **Chang, J.**, Santos J., Perez, D., Lewis, N.S., and Vincent, A.L., 2021. Evolution and antigenic advancement of N2 neuraminidase of swine influenza A viruses circulating in the United States following two separate introductions from human seasonal viruses. Journal of Virology.

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Zeller, M.A., **Chang, J.**, Vincent, A.L., Gauger, P.C. and Anderson, T.K., 2021. Spatial and temporal coevolution of N2 neuraminidase and H1 and H3 hemagglutinin genes of influenza A virus in US swine. *Virus Evolution*, 7(2), p.veab090.

Kaplan, B.S., Kimble, J.B., **Chang, J.**, Anderson, T.K., Gauger, P.C., Janas-Martindale, A., Killian, M.L., Bowman, A.S. and Vincent, A.L., 2020. Aerosol transmission from infected swine to ferrets of an H3N2 virus collected from an agricultural fair and associated with human variant infections. *Journal of Virology*.

Anderson, T.K., **Chang, J.**, Arendsee, Z.W., Venkatesh, D., Souza, C.K., Kimble, J.B., Lewis, N.S., Davis, C.T. and Vincent, A.L., 2020. "Swine Influenza A Viruses and the Tangled Relationship with Humans." *Cold Spring Harbor Perspectives in Medicine*, p.a038737.

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